HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines for ETL processes, handling approximately **5,000 records per hour** to populate analytical databases.
- Developed and maintained **Java Spring Boot** REST APIs, facilitating inter-service communication and supporting a **15% increase** in data processing throughput.
- Managed production support for enterprise applications using **Splunk** and **Dynatrace**, responding to critical incidents and maintaining service level agreements.
- Refactored complex **SQL** queries across **PostgreSQL** and **Oracle**, improving query execution times for core reporting functions by an average of **20%**.
- Containerized several microservices using **Docker**, enabling consistent deployment across development and production environments on **AWS EC2**.
- Contributed to the design and implementation of a **quantitative modelling framework**, developing core components for data ingestion and feature extraction.
- Built and deployed **machine learning models** for anomaly detection using **scikit-learn** on a dataset of **~150,000 transactions**, achieving an AUC of **0.82**.

SKILLS

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Statistical Analysis, Machine Learning

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Linux, Splunk, Dynatrace

PROJECTS

Quantitative Trading Strategy Backtester

- Developed a Python-based backtesting engine to evaluate algorithmic trading strategies across historical market data, simulating over **10 million trades**.
- Implemented statistical analysis modules in **pandas** and **NumPy** to calculate performance metrics such as Sharpe ratio, Max Drawdown, and win rates.
- Engineered a feature extraction pipeline to generate trading signals from raw market data, utilizing **SQL** for efficient data retrieval from a **PostgreSQL** data warehouse.

ML Model for Transaction Anomaly Detection

- Built a Python-based machine learning pipeline to process and transform transaction data for model training, handling a dataset of approximately **150,000 records**.
- Trained a **scikit-learn** classifier (Random Forest) to identify fraudulent transactions, achieving an Area Under the Curve (AUC) of **0.82** on a held-out test set.
- Packaged the trained model into a **FastAPI** service for real-time inference, demonstrating p95 latency under **200ms** for prediction requests.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)